

Semi-Analytical Modeling of Blade Wake Deficits for Propeller-Wing Interaction in eVTOL Aeroacoustics

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The rapid emergence of electric Vertical Take-Off and Landing (eVTOL) aircraft requires stringent control over noise levels. A key contributor to the overall acoustic signature in these configurations is the propeller-wing interaction noise, which is caused by the impingement of the unsteady wake of the propeller onto the downstream wing, and the associated modification of the wake prior to interaction. This study combines CFD-generated wake data with a tunable semi-analytical acoustic framework with the aim of offering a computationally efficient alternative to high-fidelity simulations for noise prediction. The approach models the blade wake deficit as a periodic summation of Gaussian functions, which are then used as a modulation function for the homogeneous turbulence field. This reconstructed velocity field is coupled with Amiet’s flat-plate model to determine the resulting sound pressure spectrum. The methodology employs Unsteady Reynolds Averaged Navier-Stokes (URANS) simulations of both free and installed propeller configurations to generate the wake data and investigates the influence of the wing on the reconstruction of the turbulent propeller wake. The results show that while the Gaussian-based reconstruction captures the dominant wake characteristics in free conditions, the presence of the wing introduces pronounced asymmetry, wake deformation, and vortex interaction effects that are not fully represented by the periodic formulation.

I. Introduction

DRIVEN in part by advancements in battery, distributed electric propulsion (DEP), and autonomous technologies, a new class of Urban Air Mobility (UAM) aircraft, generally referred to as electric vertical take-off and landing (eVTOL), has emerged as a potential alternative to the existing transportation system [1]. These aircraft are lightweight, do not require long runways, and are more sustainable, with fewer direct emissions [2, 3]. However, noise remains a critical concern for propeller-driven configurations, particularly due to installation effects [4, 5]. Compared to isolated propellers, installed configurations introduce additional interactional noise sources associated with the unsteady propeller wake [6–9]. In such configurations, propeller–wing interaction arises from the impingement of the evolving wake by the downstream wing, altering the incident flow field and contributing significantly to the overall acoustic signature [10].

Accurate prediction of propeller–wing interaction noise typically relies on high-fidelity methods, including computational fluid dynamics (CFD) coupled with aeroacoustic models or experimental measurements [11–13]. Capturing these interactions requires resolving the three-dimensional, unsteady wake dynamics to represent the underlying physical mechanisms accurately. While large-eddy simulation (LES) studies have provided detailed insight into wake evolution, including tip-vortex destabilization and wake diffusion due to Kelvin–Helmholtz instabilities [14, 15], their high computational cost limits their applicability in early-stage design and parametric studies.

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To address this limitation, analytical and semi-analytical models have been developed, such as adaptations of counter-rotating open rotor (CROR) theories by Akiwate et al. [6] and a lifting-line model with turbulence-corrected wake by Goates [16]. However, these approaches remain limited in their ability to capture configuration-specific effects, such as detailed blade geometry and wing installation. More advanced hybrid formulations, such as those proposed by Grasso [17], introduce parametrized wake representations for predicting interaction noise in CROR configurations. Prior studies suggest that viscous wake–interaction and tip-vortex interaction noise dominate the total interaction noise, while potential interactions become significant only at small rotor–wing separations [6]. Accurate modeling of the viscous wake is therefore critical for predicting the resulting noise spectrum.

In response to these challenges, a hybrid methodology is proposed that combines CFD-derived wake fields with a semi-analytical acoustic framework based on Amiet’s leading-edge noise theory [18]. Following the analytical wake modeling strategy of Grasso [17], originally developed for CROR configurations, the approach is adapted here to account for propeller-wing interaction effects in eVTOL systems. Wake deficit profiles extracted from unsteady Reynolds-averaged Navier–Stokes (URANS) simulations are represented using a tunable Gaussian formulation that captures the dominant coherent structures. The reconstructed velocity field is then used as input to the acoustic model to predict leading-edge noise induced by wake–wing interaction. This formulation provides a computationally efficient alternative to high-fidelity simulations, making it particularly suitable for early-stage eVTOL design exploration.

II. Acoustic Formulation and Wake Modeling Framework

The acoustic emissions of a propeller in an installed configuration differ from those of an isolated propeller operating in free air due to aerodynamic interactions with surrounding surfaces [19–22]. In propeller–wing configurations, the total radiated sound is commonly decomposed into isolated propeller noise and interaction noise [6]. While the tonal and broadband characteristics of isolated propellers in uniform inflow are well established [7, 23, 24], the characterization and prediction of interaction noise remain less established for emerging eVTOL configurations, motivating further investigation.

A. Leading-Edge Noise Model for Turbulent Inflow

The corresponding wake-interaction noise is modeled using Amiet’s theory for sound generation at the leading edge of an airfoil immersed in a homogeneous turbulent flow [18]. Amiet’s theory assumes a uniform flow field and an airfoil that is infinitely thin, camberless, and operating at a zero angle of attack [18, 25, 26]. The far-field power spectral density (PSD) of the acoustic pressure for an airfoil with a large aspect ratio is expressed as

$$S_{pp}(\mathbf{x}, \omega) = \left(\frac{\rho_0 k x c}{2\sigma_0^2} \right)^2 \pi W \frac{d}{2} \varphi_{vv}(K_z, K_x) |\mathcal{L}(\mathbf{x}, K_z, K_x)|^2, \quad (1)$$

where $\mathbf{x} = (x, y, z)$ denotes the position in the reference frame as illustrated in Fig. 1, and $\sigma_0 = \sqrt{z^2 + \beta^2(x^2 + y^2)}$ is the observer radius corrected for convection effects, with $\beta = \sqrt{1 - M_\infty^2}$, where M_∞ is the freestream Mach number. The term φ_{vv} represents the two-dimensional wavenumber spectrum of the wall-normal velocity, commonly referred to as the upwash velocity. The function $\mathcal{L}$, given in section A, denotes the aerodynamic transfer function, which consists of the leading-edge contribution $\mathcal{L}_1$ and the trailing-edge contribution $\mathcal{L}_2$, corrected for back-scattering effects [27, 28]. Each emitted frequency ω corresponds to a specific value of the chordwise aerodynamic wavenumber, defined as $K_z = \omega/W$, where W denotes the convection velocity. Since the wing is assumed to have a large aspect ratio, the far-field observer perceives only the gusts that generate an acoustic wavefront normal to the line connecting the airfoil and the observer. Consequently, the spanwise wavenumber $K_x = \omega x / (c_0 \sigma_0)$ depends on the observer radius. Recent extensions of Amiet’s formulation have also been proposed to account for more realistic airfoil geometries and flow conditions, and may be incorporated in future developments of the present framework [29, 30].

B. Semi-Analytical Representation of Periodic Blade Wake Deficits

To evaluate the acoustic response using Amiet’s formulation, the incoming upwash velocity spectrum at the leading edge must be specified. In the present work, this spectrum is constructed by representing the propeller wake as a periodic modulation of an underlying turbulent field. This formulation assumes that the incoming flow can be decomposed into a deterministic mean wake deficit and superimposed turbulent fluctuations, such that the mean component governs the tonal response while the turbulent fluctuations contribute to the broadband component [17]. Within this linearized

framework, the wake is modeled as the product of a two-dimensional wavenumber-resolved turbulent velocity component and a modulation function describing the periodic mean velocity deficit.

$$v(t) = \underbrace{v'(t)}_{\text{Turbulent}} \times \underbrace{f(t)}_{\text{Modulation function}}. \quad (2)$$

The modulation function must reproduce the mean blade-wake deficit while preserving periodicity across successive blade passages. A Gaussian profile is adopted to represent the temporal evolution of an individual wake crossing at the airfoil leading edge. The resulting normalized wake kernel is written as

$$h(t) = \exp\left(-\frac{t^2}{2\sigma_t^2}\right), \quad (3)$$

where σ_t denotes the standard deviation controlling the temporal extent of a single wake.

A periodic train of blade wakes with a period of $T = 2\pi/B\Omega$ is obtained by repeating the single Gaussian kernel over successive blade passages, yielding

$$f(t) = \sum_{n=-\infty}^{\infty} h(t - nT) = \sum_{n=-\infty}^{\infty} \exp\left(-\frac{(t - nT)^2}{2\sigma_t^2}\right). \quad (4)$$

This representation assumes that adjacent blade wakes remain sufficiently distinct so that their individual deficits can be treated as non-overlapping perturbations [28]. Applying Poisson's summation formula yields the corresponding Fourier-series expansion:

$$f(t) = \frac{1}{T} \sum_{\lambda=-\infty}^{\infty} H_{\lambda} \exp\left(\frac{i2\pi\lambda t}{T}\right), \quad (5)$$

where H_{λ} denotes the Fourier coefficients of the Gaussian wake train.

$$H_{\lambda} = \sigma_t \sqrt{2\pi} \exp\left(-\frac{2\pi^2\sigma_t^2\lambda^2}{T^2}\right). \quad (6)$$

In this representation, λ denotes the harmonic index associated with integer multiples of the blade-passing frequency $1/T$, with $\lambda = 0$ corresponding to the mean component. For a Gaussian wake, the harmonic amplitudes decay exponentially with increasing λ , as governed by the factor $\exp[-2\pi^2\sigma_t^2\lambda^2/T^2]$. Consequently, narrower wakes generate broader harmonic content, whereas wider wakes concentrate energy within the lowest harmonics.

Taylor's frozen-wake hypothesis is applied by assuming that the wake deficit convects past the leading edge at velocity W without deformation [31]. The disturbance field is therefore written in convected coordinates $w(z/W - t, t)$, allowing the temporal wake train to be interpreted as a spatially transported structure, consistent with Amiet's frozen-turbulence assumption [18].

The periodic wake is modeled as a superposition of oblique sinusoidal gusts. The temporal phase maps to the streamwise wavenumber $\kappa_z = 2\pi/(WT)$, while wake inclination introduces a spanwise wavenumber κ_x . Assuming that the wake wavefront intersects the leading edge at a constant angle γ , the two-dimensional modulation is constrained by $\tan \gamma = \kappa_x/\kappa_z$ defined in Fig. 1, yielding the generalized two-dimensional modulation function

$$f(z, x, t) = \sum_{\lambda=-\infty}^{\infty} H_{\lambda} \exp\left(i\lambda\left(\kappa_z z + \kappa_x x - \frac{2\pi t}{T}\right)\right). \quad (7)$$

The wavenumber κ_x can be defined $\kappa_x = 2\pi/R_1$, where R_1 is a fictional spanwise period as suggested by Carazo et al. [32], leading to a three-dimensional formulation derived in section B.

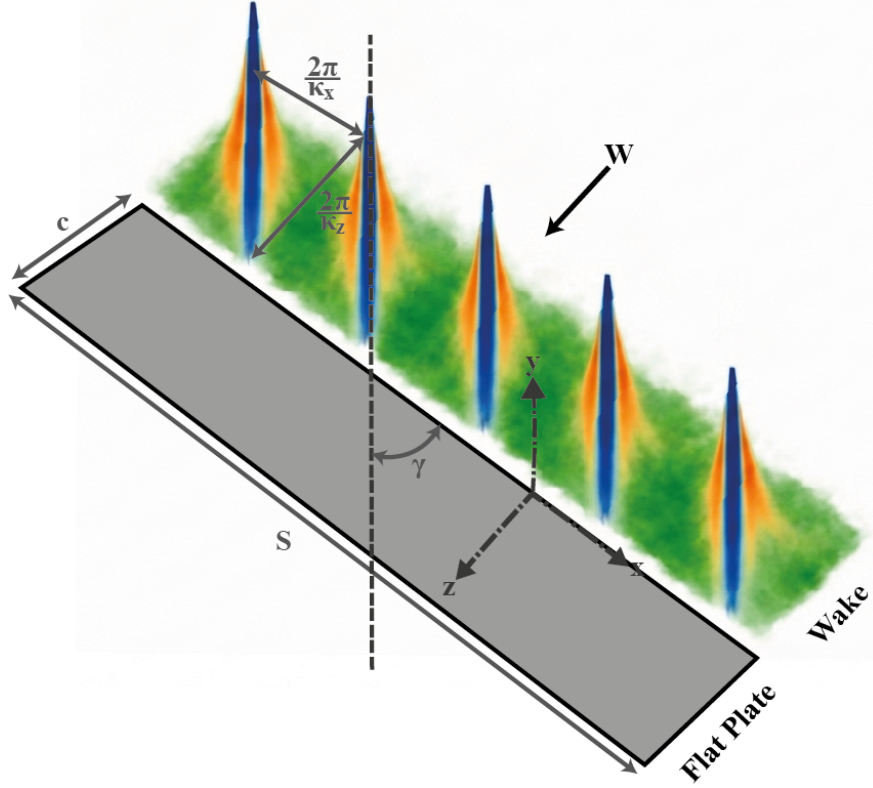


Fig. 1 Schematic of a turbulent flow convecting with velocity W toward a flat plate. The inclined wavefront intersects the leading edge at angle γ , defining the spanwise and streamwise wavenumber components.

Substituting Eq. (7) into Eq. (2) and applying the double spatial Fourier transform ($\hat{\cdot}$) in the streamwise and spanwise directions gives

$$\begin{aligned}\hat{v}(k_z, k_x, t) &= \frac{1}{(2\pi)^2} \sum_{\lambda=-\infty}^{\infty} H_{\lambda} \exp\left(-\frac{i2\pi\lambda t}{T}\right) \iint_{-\infty}^{\infty} v'(z, x, t) \exp(i\lambda(\kappa_z z + \kappa_x x)) \exp(-i(k_z z + k_x x)) dz dx \\ &= \sum_{\lambda=-\infty}^{\infty} H_{\lambda} \exp\left(-\frac{i2\pi\lambda t}{T}\right) \hat{v}'(k_z - \lambda\kappa_z, k_x - \lambda\kappa_x).\end{aligned}\quad (8)$$

The wake modulation redistributes the turbulent energy in wavenumber space through shifts governed by the harmonic index λ as described by Eq. (8). Hence, the corresponding energy spectral density of the wall-normal velocity fluctuations is therefore obtained by summing the contributions of all shifted components, yielding the broadband velocity spectrum

$$\varphi_{vv}(k_z, k_x) = \sum_{\lambda=-\infty}^{\infty} |H_{\lambda}|^2 \varphi_{v'v'}(k_z - \lambda\kappa_z, k_x - \lambda\kappa_x), \quad (9)$$

Substituting Eq. (9) into Amiet's leading-edge theory Eq. (1) directly provides the broadband acoustic response

$$S_{pp}(\mathbf{x}, \omega) = \left(\frac{\rho_0 k x c}{2\sigma_0^2}\right)^2 \pi W \frac{d}{2} |\mathcal{L}(\mathbf{x}, K_z, K_x)|^2 \sum_{\lambda=-\infty}^{\infty} |H_{\lambda}|^2 \varphi_{v'v'}(K_z - \lambda\kappa_z, K_x - \lambda\kappa_x). \quad (10)$$

The tonal contribution is recovered from the same formulation by setting $v'(t) = 0$, such that $v(t) = f(t)$ and the wavenumber spectrum reduces to discrete harmonic contributions. These are represented by Dirac distributions centered at integer multiples of the fundamental wake wavenumbers, replacing the continuous turbulence spectrum $\varphi_{v'v'}$ in Eq. (10) and evaluating the tonal contribution within the same Amiet's leading-edge theory as a set of coherent gusts at blade-passing harmonics.

III. Numerical and Data-Driven Methodology

A. URANS Simulation Setup and Numerical Configuration

A detailed understanding of the flow behavior in the wake downstream of a propeller is essential to capture the underlying physics and predict acoustic responses. URANS simulations are conducted using the commercial software package Star-CCM+. A coupled flow solver simultaneously solves the mass, momentum, and energy equations, ensuring that the velocity–pressure coupling is handled adequately. The wall-resolved SST $k - \omega$ turbulence model is employed to capture both near-wall and free-stream flow features. The 6DOF rigid body equations of motion are not solved, as the propeller motion is prescribed to rotate at a constant speed of $n = 5000$ rpm.

1. Geometry Definition and Computational Domain Setup

The propeller implemented in the simulation is designed to represent the forward-facing propellers of a representative eVTOL aircraft, following the configuration described by Barker et al. [33], as shown in Fig. 2. A Cartesian coordinate system is employed, with the spanwise direction defined as X , the normal direction as Y , and the streamwise/chordwise direction as Z , aligned with the convection speed W . While the full-scale propellers have a diameter of 3.04 m and a hub diameter of 0.46 m, a geometrically scaled model, reduced by a factor of ten, is used to decrease computational cost and match the experimental hovering conditions [34]. In addition to the isolated propeller case, a winged configuration incorporating a NACA 4418 airfoil with a chord length of 1.70 m, positioned 2.42 m downstream of the propeller plane, is created. To maintain the same advance ratio, $J = W_\infty/nD$, the freestream velocity is set to 30 m s^{-1} , resulting in $J = 0.88$.

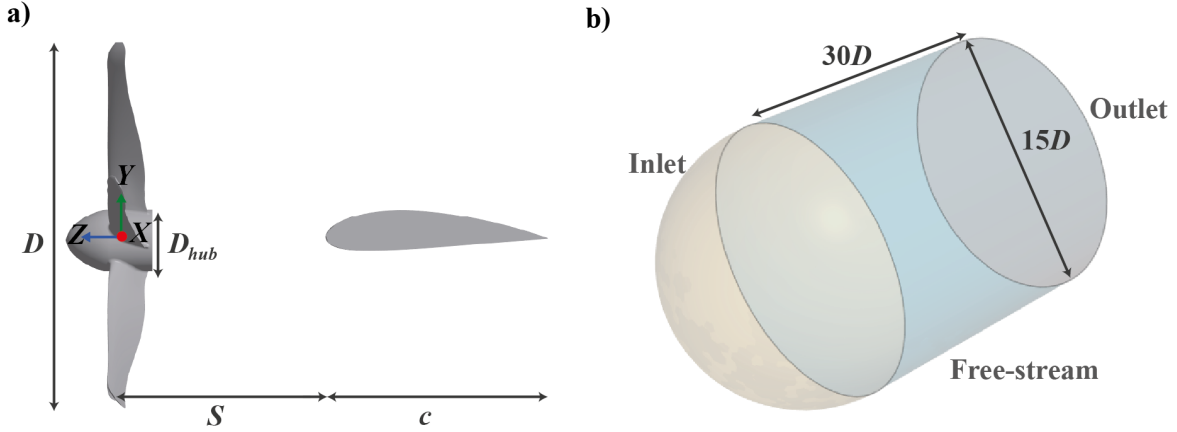


Fig. 2 Geometry and computational setup for the URANS simulations: (a) scaled propeller–wing configuration, and (b) cylindrical computational domain showing inlet, outlet, and far-field boundaries.

The computational domain has a cylindrical shape extending over $0 < Z < 30D$, with a spherical inlet of diameter $15D$, and the propeller is positioned at $(X, Y, Z) = (0, 0, 0)$. The boundary conditions consist of a velocity inlet at the upstream boundary, a zero-gradient Neumann condition at the outlet, a no-slip wall condition on the propeller surfaces, and a slip condition on the outer cylindrical boundary.

2. Grid Generation and Resolution Strategy

A hybrid computational grid composed of polyhedral control volumes and prismatic near-wall layers is employed to resolve the propeller slipstream and its interaction with the wing surface while limiting numerical dissipation. The final mesh contains approximately 7.83×10^6 cells distributed over three regions: the rotating propeller region (2.38×10^6 cells), the stationary near-field wing region (2.60×10^6 cells), and the outer far-field domain (2.85×10^6 cells). The sliding mesh method is employed so that the two regions meet at an interface without overlapping and allow flow information to be transferred across the interface via interpolation at every time step.

The propeller surface is discretized using approximately 500 000 faces, corresponding to an effective surface resolution of approximately 420 chordwise and 210 spanwise divisions. Near solid surfaces, 16 prism layers are

generated with a first layer height of 1.5×10^{-5} m, resulting in $Y^+ \approx 1$. Additional local refinement is applied in the propeller wake and wing interaction region to improve the resolution of tip vortices, wake contraction, and boundary-layer development. A fixed time step of $\Delta t = 3.33 \times 10^{-5}$ s is employed, corresponding to approximately 1° of propeller rotation per time step.

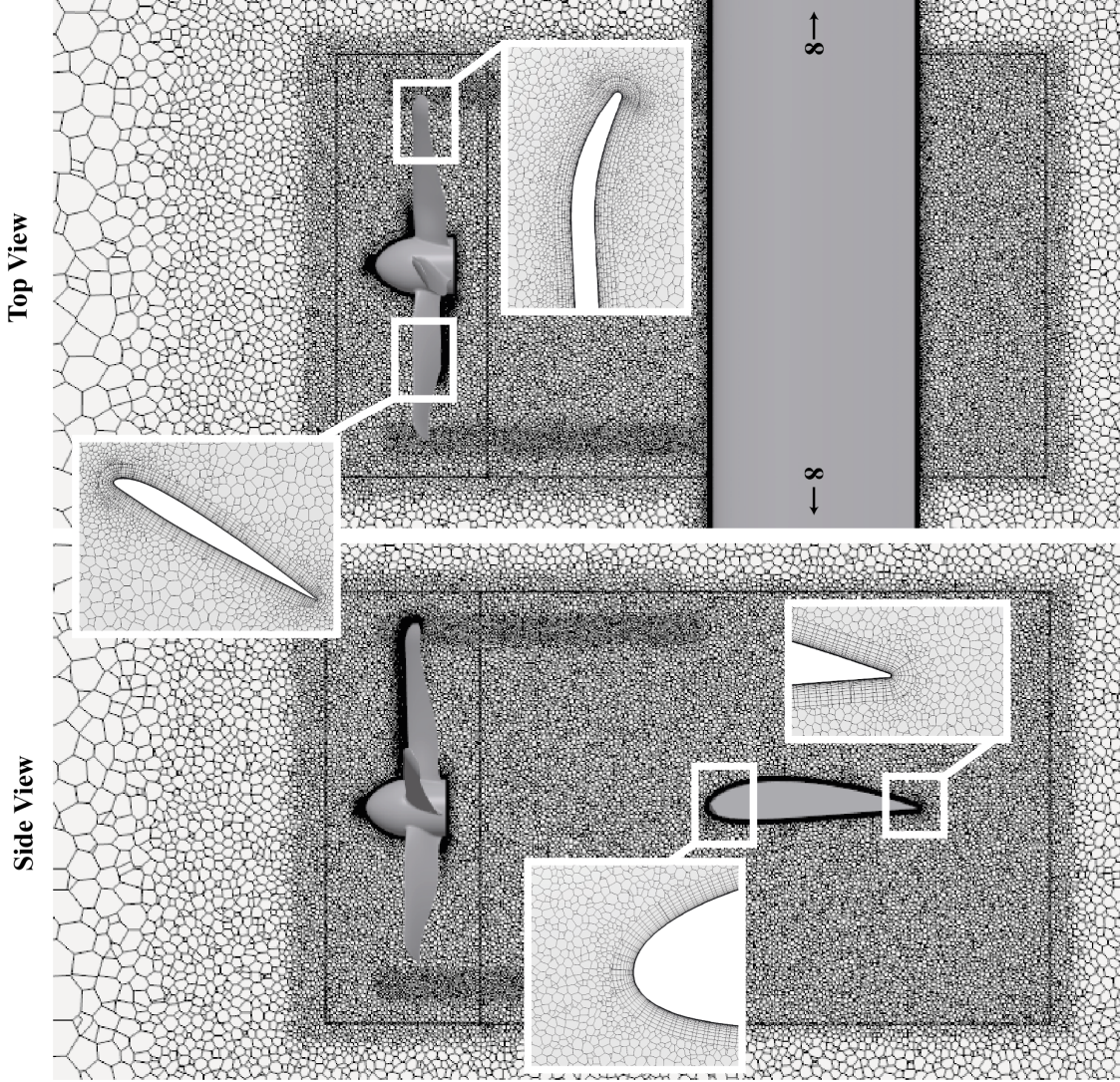


Fig. 3 Illustration of computational mesh for URANS propeller-wing configuration.

3. Aerodynamic and Acoustic Validation Strategy

The numerical framework is further validated against isolated propeller measurements obtained in the von Karman Institute for Fluid Dynamics (VKI) ALCOVES anechoic chamber [34]. As experimental data for the present propeller configuration are only available under hovering conditions, the validation is conducted in hover rather than in cruise. Acoustic measurements were performed using three linear microphone arrays comprising 24 G.R.A.S. 40PL condenser microphones distributed around the propeller [34] as shown in Fig. 6a. Starting from the cruise URANS configuration, the hovering case is obtained by modifying the blade twist distribution, inlet velocity, rotational speed (8300 rpm), and time step, while preserving all remaining numerical settings. The temporal discretization is selected to maintain an angular resolution of 1° per time step.

The corresponding results are presented in section IV, while a sensitivity analysis of the results to the turbulence modeling strategy will be assessed in subsequent analyses using LES simulations, in order to quantify its influence on wake diffusion, particularly with respect to the decay and spatial spreading of the blade wake deficit.

B. Wake Deficit Extraction and Parametric Characterization

The modulation function described by Eq. (2) represents the blade wake deficit in the direction perpendicular to the local motion of the blade. The procedure used to extract this wake deficit is illustrated in Fig. 4. First, the CFD flow field is interpolated onto a structured cylindrical grid defined by (N_r, N_θ, N_Z) , with N_r representing the number of radial strips. The cylinder length is prescribed by the wing location (panel a), after which the field is partitioned into thin cylindrical sections in the radial direction (panel b). Each radial strip is then unwrapped onto a two-dimensional plane defined by the streamwise coordinate Z/S and the circumferential coordinate $r\theta/D$ (panel c). The unwrapped planes are centered at their corresponding blade trailing edge location to account for radial variations in blade camber. A linear fit of the blade wake deficit is then obtained using the Random Sample Consensus (RANSAC) method [35] while excluding neighboring blade wakes from the fit (panel d).

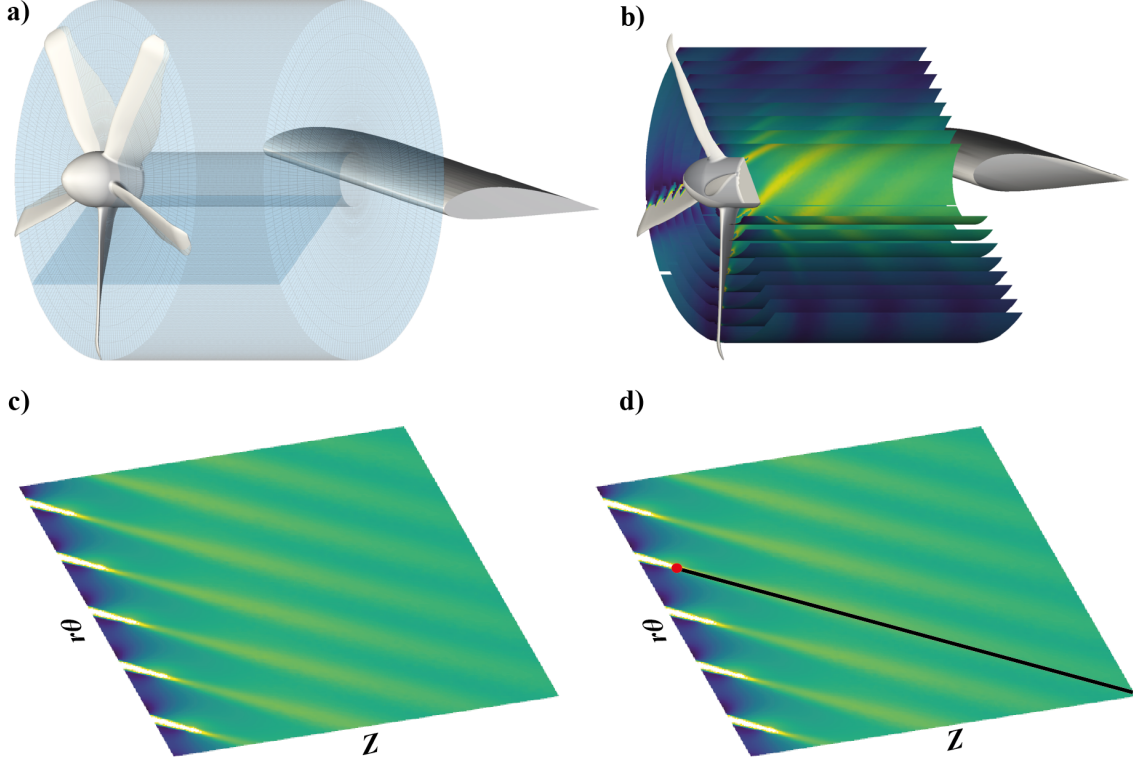


Fig. 4 Wake extraction methodology from URANS data: (a) cylindrical interpolation domain, (b) radial slicing of the flow field, (c) unwrapped circumferential velocity field, and (d) identification of wake centerline using RANSAC fitting, showing the trailing edge point (red sphere) and the line of best fit along the wake (black line).

Although the analytical formulation introduced in section II is expressed in the temporal domain through the wake passage time t , direct extraction from CFD is more naturally performed in local wake-normal coordinates (η, ζ) . Under Taylor’s frozen-wake assumption, temporal and spatial descriptions remain equivalent through $\eta = Wt$, where W denotes the local convection velocity. For each radial strip, the wake is treated locally at a fixed radius r , such that the circumferential periodicity of successive blade wakes is governed by the local blade pitch $P = 2\pi r/B$, which also satisfies $P = WT$. The wake deficit is therefore fitted in spatial form using a Gaussian profile:

$$h(\eta) = A \exp\left(-\frac{\eta^2}{2b^2}\right), \quad (11)$$

where the amplitude A and width $b = W\sigma_t$ vary with radial position and downstream distance according to the extracted local wake profile.

Within each strip, the circumferential direction $s = r\theta$ is treated locally as a tangential coordinate, consistent with the strip-wise approximation used in annular wake models [17, 27, 32]. Because the wake intersects this local tangential

direction at an angle γ , the Gaussian width is projected as $b_\theta = b/\cos \gamma$, as illustrated in Fig. 5. The corresponding periodic wake modulation field for each strip is then written as

$$\Delta w(r, \theta, Z) = \frac{A}{\sigma_\lambda \sqrt{2\pi}} \sum_{\lambda=-\infty}^{+\infty} \exp\left(-\frac{1}{2} \left(\frac{\lambda}{\sigma_\lambda}\right)^2\right) \exp\left(i\lambda B \left(\theta + \frac{s_0}{r} + \frac{Z \tan \gamma}{r}\right)\right), \quad (12)$$

where $\sigma_\lambda = r/(Bb_\theta)$. Dividing $\Delta w(r, \theta, Z)$ by the local amplitude A recovers the unit-amplitude modulation function of Eq. (2). The term s_0 represents the circumferential displacement of the extracted wake center relative to the local blade trailing-edge reference, while $Z \tan(\gamma)/r$ accounts for the phase shift due to downstream convection of an inclined wake. At mid-span, the wake remains quasi-two-dimensional, with the velocity deficit aligned with the trailing edge. Near the tip, however, rapid roll-up of the trailing vortex sheet produces strong three-dimensional effects and transverse velocities that reorient the flow, advecting the deficit toward the suction side [36]. This displacement is consistent with observed wake transport behavior [37], and is represented in the present formulation through the offset s_0 , obtained from CFD-based wake extraction.

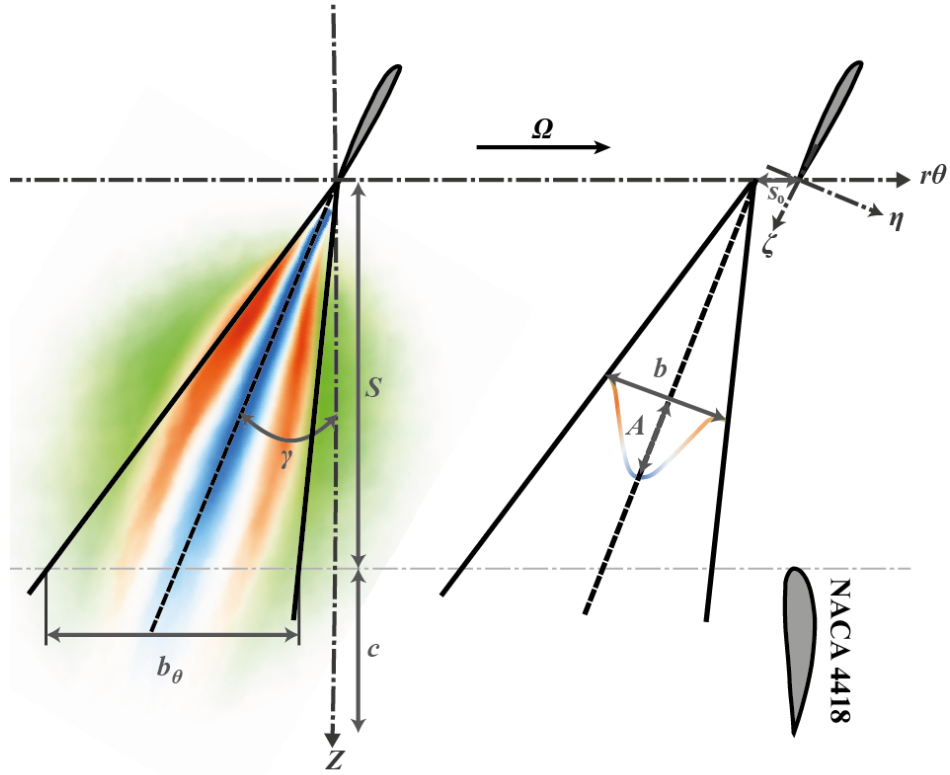


Fig. 5 Gaussian representation of the blade wake deficit showing model parameters, adapted from [17].

The wake modulation is evaluated independently at each radial strip using the locally extracted parameters $(A, b_\theta, s_0, \gamma)$, which are specific to the wake generated by an individual blade passage. Parameter identification is performed within the downstream interval $0.2 \leq \zeta \leq 0.8$, thereby avoiding regions where pronounced three-dimensional propeller effects and wing-induced blockage compromise the validity of the Gaussian representation. For each strip, the analytical reconstruction preserves the harmonic structure of the formulation through the local blade-passage wavenumbers $k_s = B/r$ and $k_r = k_s \tan(\gamma)$, which define the circumferential and streamwise spectral content associated with the inclined wake deficit.

Under the frozen-wake assumption, this strip-wise treatment preserves the dominant local wavenumber content required by the analytical response model and allows the extracted blade wake deficit to be interpreted as an oblique gust convecting toward the wing leading edge.

IV. Results

A. Validation Against Hovering Experimental Data

Table 1 compares the predicted and measured aerodynamic coefficients. Deviations remain below 5% for both thrust and power coefficients, indicating that the numerical setup accurately reproduces the dominant aerodynamic behavior of the propeller under hover conditions.

Quantity	EXP [34]	CFD	Dev. (%)
C_T [-]	0.231	0.222	-4.04
C_P [-]	0.117	0.120	+3.03

Table 1 Comparison between experimental and URANS aerodynamic coefficients $C_T = T/(\rho n^2 D^4)$, $C_P = P/(\rho n^3 D^5)$ for the hovering isolated propeller case.

The CFD acoustic prediction was computed using the VKI in-house aeroacoustic solver BATMAN^π, based on the instantaneous blade surface pressure. The blade surface was discretized into 10 radial strips and 2 azimuthal sectors per blade, and the loading noise was reconstructed through a fixed dipole formulation using 300 dipoles. Listeners were positioned on a circular arc of radius $R = 1$ m, with an angular discretization of 5° , as shown in Fig. 6b. The resulting experimental and CFD data were subsequently normalized with respect to propagation distance.

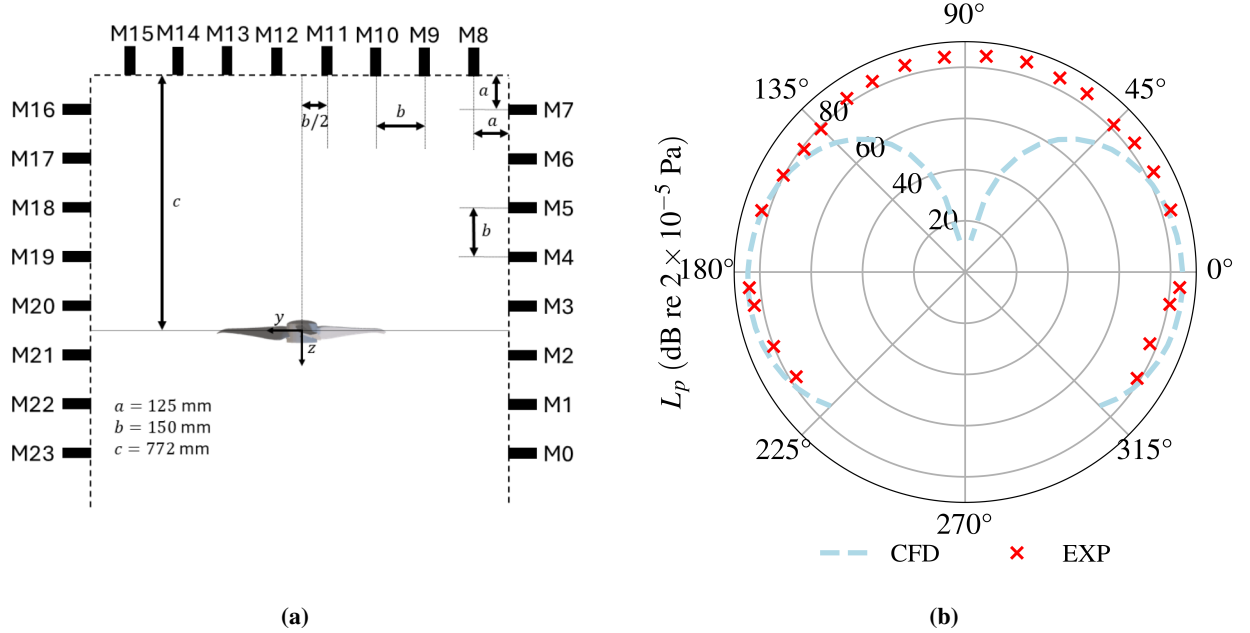


Fig. 6 (a) Microphone layout in the ALCOVES anechoic chamber [34], discussed in section III and (b) experimental and numerical directivity patterns of the far-field noise radiated by the isolated propeller at the BPF, observer plane $\phi = 90^\circ$.

Results indicate that the numerical framework reproduces the measured directivity satisfactorily in the vicinity of the rotor plane, where tonal loading dominates the acoustic field. Larger discrepancies appear near the propeller axis, where the numerical model underpredicts the radiated sound pressure level. This discrepancy may originate from additional unsteady loading mechanisms not represented in the current numerical framework, including possible flow recirculation effects within the closed anechoic chamber, as reported in previous studies [38, 39]. Despite these limitations, the agreement obtained in the dominant radiation region confirms that the present framework provides an adequate aerodynamic-acoustic basis for installation-noise analyses.

B. Propeller Wake Vortex Structure and Evolution

The instantaneous vortex structures based on visualizing the flow field of the installed propeller-wing configuration are shown in Fig. 7. It illustrates where the propeller tip vortices are generated and convected, their interaction with the wing, and their dissipation further downstream.

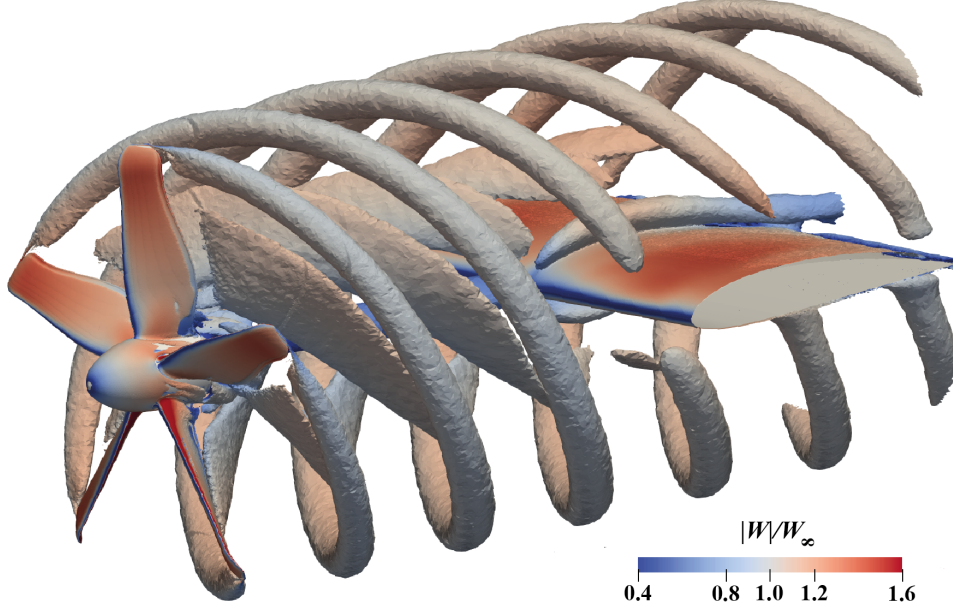


Fig. 7 Instantaneous vortex structures in the installed configuration identified using the non-dimensional Q-criterion $\bar{Q} = 0.15$ [40], and colored by the velocity magnitude.

The following coherent wake structures are identified: tip vortices forming thin helical tubes at the blade tips, blade wakes originating near mid-span as thicker helical filaments, and hub vortices exhibiting oscillatory motion associated with low velocity regions near the hub. A visual comparison between the free and installed configurations shown in Fig. 8 indicated that, up to the wing leading edge, the wake trajectories remain nearly identical in both configurations.

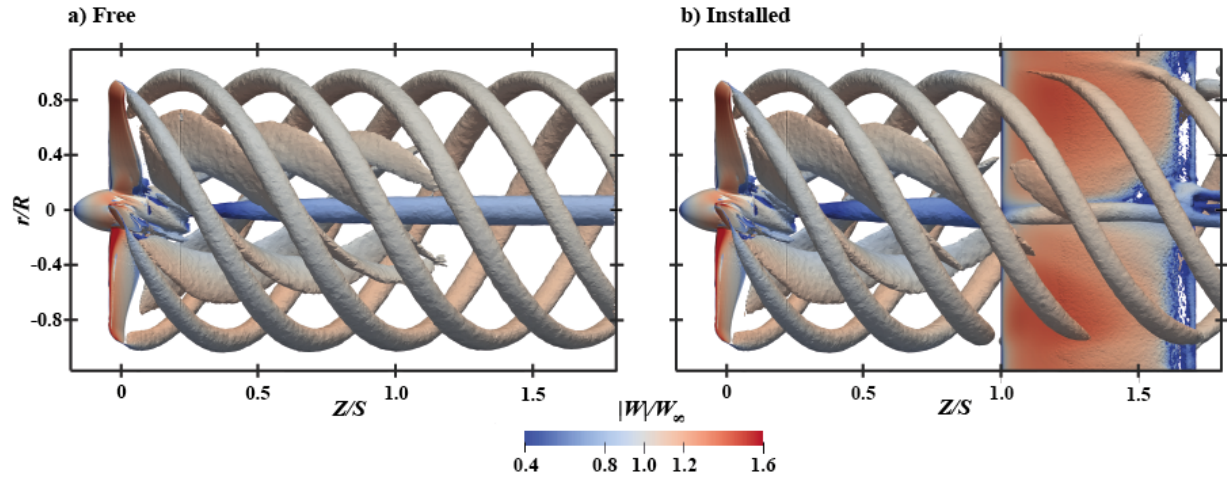


Fig. 8 Top-view visualization of the propeller wake vortex structures identified using the non-dimensional Q-criterion $\bar{Q} = 0.15$ [40], and colored by the velocity magnitude.

Once the wake reaches the wing, the tip vortex is deflected in the direction of propeller rotation, in agreement with previous numerical and experimental observations [41, 42]. The rotating slipstream induces an asymmetric velocity field

over the wing, generating local downwash on the advancing side and upwash on the retreating side, thereby modifying the effective leading-edge angle of attack.

To highlight the impact of the wing on the wake evolution, Fig. 9 presents the turbulence intensity downstream of the propeller, defined as $TI = \sqrt{\frac{2}{3}}k/W_\infty$, where k denotes the turbulent kinetic energy. Three downstream locations are considered for both the free configuration and the installed configuration: immediately downstream of the propeller at $Z/S = 0.3$, directly upstream of the wing leading edge at $Z/S = 1.0$, and downstream of the wing trailing edge at $Z/(S+c) = 1.0$. At the first location, shown in (a) and (d), no significant differences are observed between the two configurations. Immediately upstream of the wing leading edge, the wake center becomes slightly asymmetric in the installed case due to the presence of the wing, while the tip vortices and blade wake remain largely unaffected. Downstream of the wing, however, a clear displacement of both the tip vortices and blade wake is observed in the direction of propeller rotation.

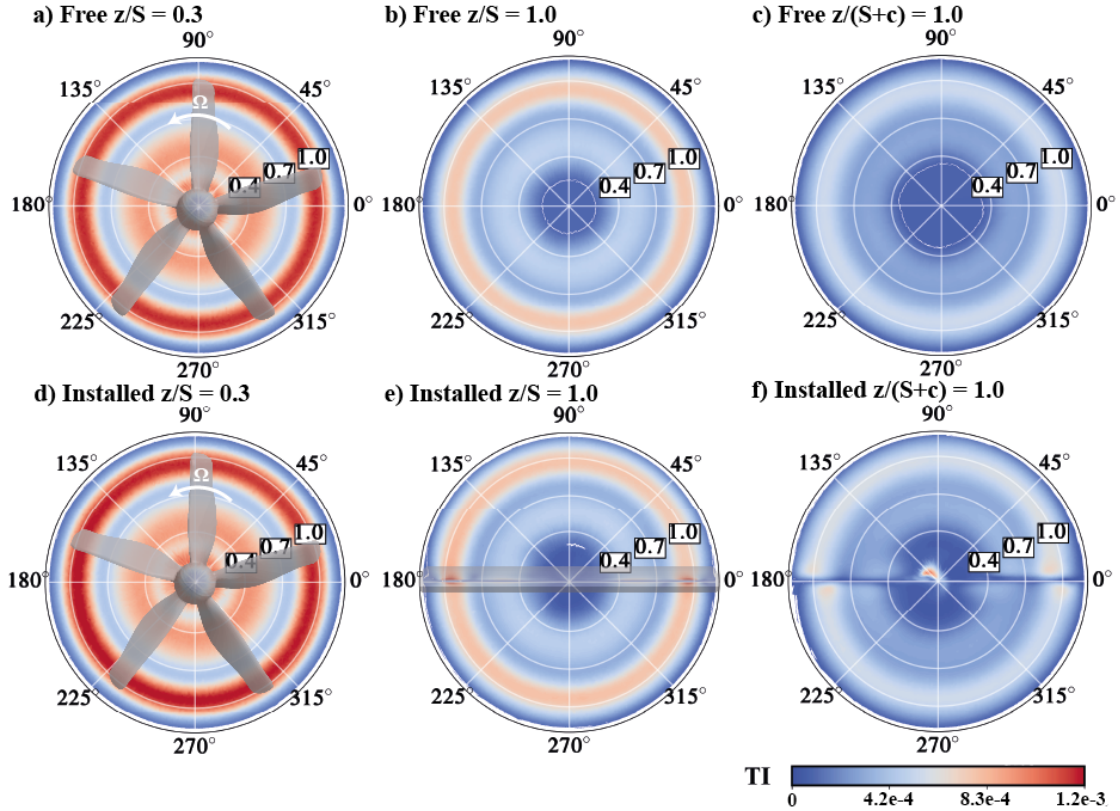


Fig. 9 Radial distribution of turbulence intensity $TI = \sqrt{\frac{2}{3}}k/W_\infty$ at selected downstream locations for free and installed configurations.

C. Wing Surface Pressure Response to Wake Interaction

To assess the effect of the propeller slipstream on the downstream wing, the instantaneous pressure fluctuations and time-averaged pressure coefficient distributions are shown in Fig. 10. The instantaneous pressure fields reveal streamwise elongated structures with a diagonal orientation, indicating the advection of helical wake features under the combination of axial convection and swirl. The nature of these patterns suggests that the dominant unsteady loading is governed by coherent blade wake convection, which gives rise to phase-locked pressure fluctuations associated with the tonal component of the interaction noise, consistent with the observations of Comunian et al. [43]. Superimposed on these large-scale structures, localized regions of elevated pressure fluctuations are observed near the outer spanwise locations, which can be attributed to the impingement of tip vortices. While these features are initially coherent, their progressive distortion and spatial spreading downstream indicate a loss of temporal coherence due to interaction with the wing boundary layer. This mechanism promotes a transfer of energy from coherent to stochastic pressure

fluctuations, leading to high-frequency content characteristic of broadband noise generation. The combined presence of wake-induced forcing and localized vortex interactions indicates that the acoustic response arises from a coupled mechanism, rather than a purely tonal or broadband source. Consequently, the resulting spanwise non-uniform loading implies a spatially distributed acoustic source, with peak contributions aligned with the regions of strongest wake and vortex interaction. The time-averaged pressure coefficient in Fig. 10(c) further shows increased suction levels in the outboard section ($r/R = 0.92$), particularly near the leading edge, indicating that the slipstream enhances the local aerodynamic loading and reinforces the non-uniformity of the acoustic source distribution along the span.

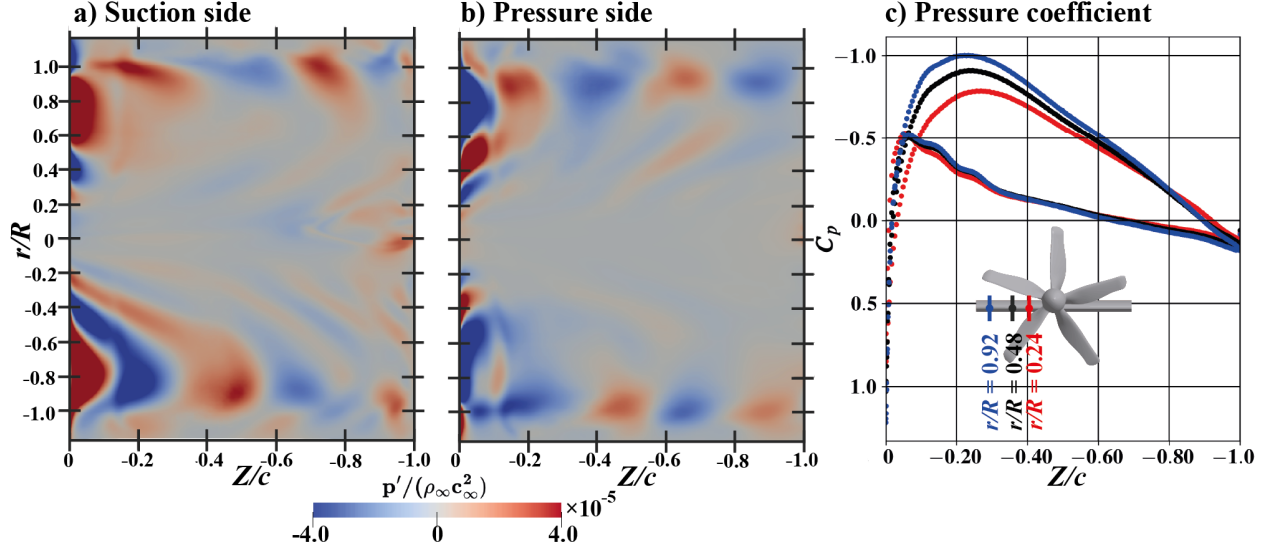


Fig. 10 (a,b) Instantaneous hydrodynamic pressure fluctuation on the wing. (c) Pressure coefficient on the installed wing.

D. Evaluation of Gaussian Wake Reconstruction

The Gaussian fit to the extracted wake profiles at representative downstream and radial locations is shown in Fig. 11a.

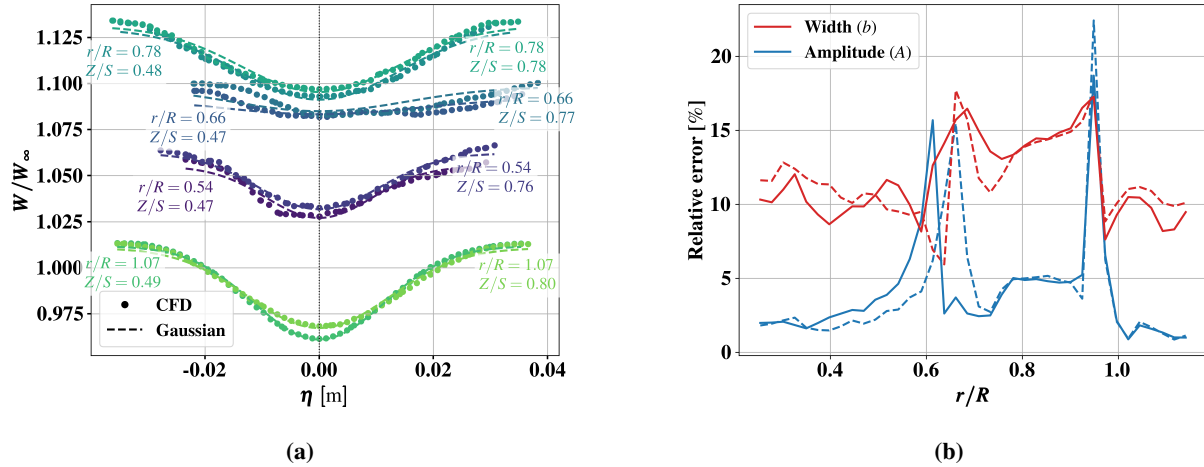


Fig. 11 (a) Comparison between the CFD-derived wake profile and the corresponding Gaussian fit for installed configuration (b) Relative fitting error, where the solid line denotes the installed configuration and the dashed line the free configuration.

The agreement is generally good near the core of the Gaussian profile, although the fit does not fully capture the profile edges, indicating that the wake deficit is not strictly Gaussian. In the blade-wake region, approximately

$0.30 < r/R < 0.65$, the deficit remains narrow with a moderate velocity reduction that weakens downstream. Closer to the tip, approximately $0.70 < r/R < 0.95$, the vortex produces a stronger velocity deficit together with elevated edge velocities, indicating sharper local velocity gradients. A measurable wake deficit also persists beyond the propeller radius, up to $r/R \approx 1.15$, after which the velocity field approaches uniform conditions.

The corresponding fitting error is presented in Fig. 11b. The mean amplitude error remains below 5%, whereas the width error is approximately 14%, primarily due to the inability of a single Gaussian function to capture the profile shoulders observed in Fig. 11a. Two distinct peaks are evident in both error metrics: the first corresponds to the transition from the blade-wake region to the tip-vortex region, while the second marks the transition from the tip vortex to the outer slipstream. The wake profile at $r/R = 0.66$, shown in Fig. 11a, lies within this transitional region and exhibits a broadened, asymmetric shape for which a clear Gaussian peak is not well defined, leading to a degradation in fitting accuracy.

The velocity deficit for each radial slice, obtained from the URANS simulations and reconstructed using the Gaussian wake model defined in Eq. (12), is presented for the free configuration at representative downstream stations $Z/S = 0.2$, 0.4, and 0.8, as shown in Fig. 12. The black lines indicate radial locations where the velocity field is reconstructed via interpolation from adjacent slices, corresponding to regions previously identified as exhibiting elevated fitting errors. The Gaussian model reproduces the velocity deficit with good agreement in both amplitude and radial location. A systematic but modest underprediction of the wake width is observed, consistent with the limitations of a single-Gaussian representation noted earlier. Nevertheless, this discrepancy remains localized and does not significantly alter the overall wake topology.

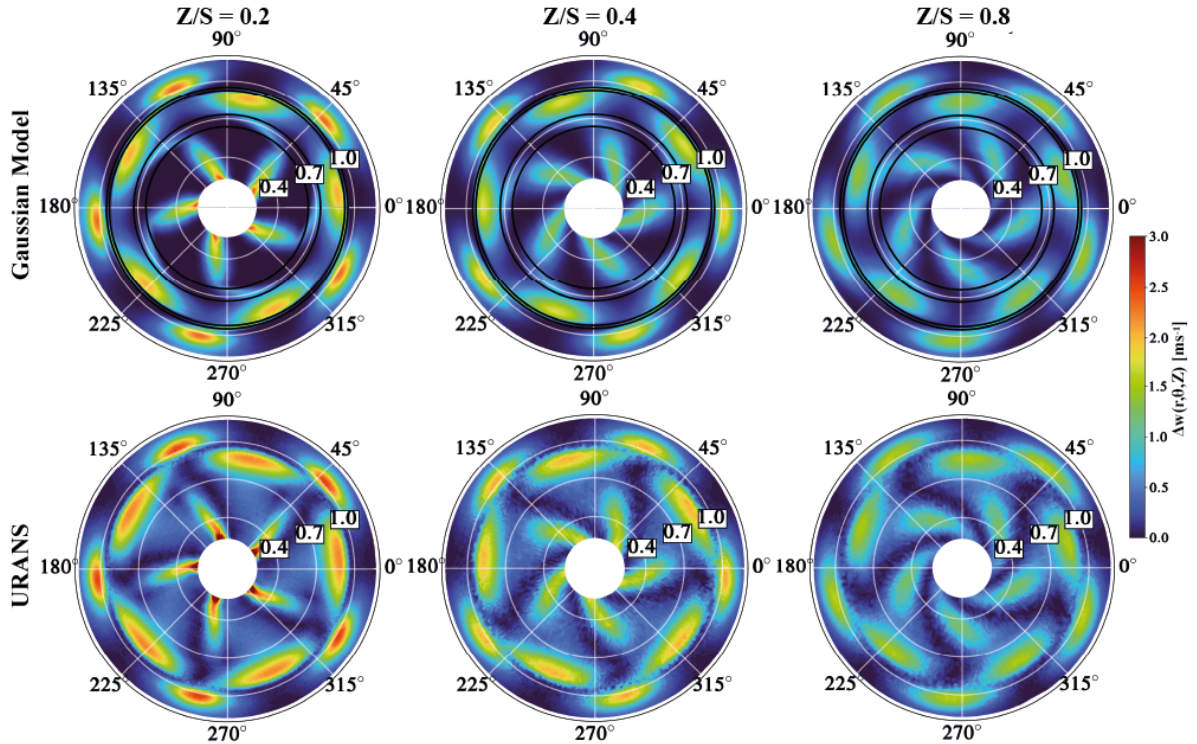


Fig. 12 URANS and Gaussian model velocity deficit for the free conditions at $Z/S = 0.2, 0.4$, and 0.8 , with black lines indicating interpolated region bounds.

In contrast, notable deviations arise for the installed configuration shown in Fig. 13. The URANS data exhibit a pronounced asymmetry, with velocity deficits up to 30% on the pressure side of the wing compared to those predicted by the model. The shape of the velocity deficit is also substantially altered by the presence of the wing: the vortices undergo deformation, begin to interact, and progressively merge, leading to a loss of distinct coherent structures. This effect becomes more pronounced near the leading edge ($Z/S = 0.8$), where strong blockage effects are not captured.

These discrepancies originate from the strict periodicity enforced by the Gaussian formulation, which reconstructs the wake as a repetition of a single blade trailing-edge deficit characterized by (A, b) . The assumption holds in the free

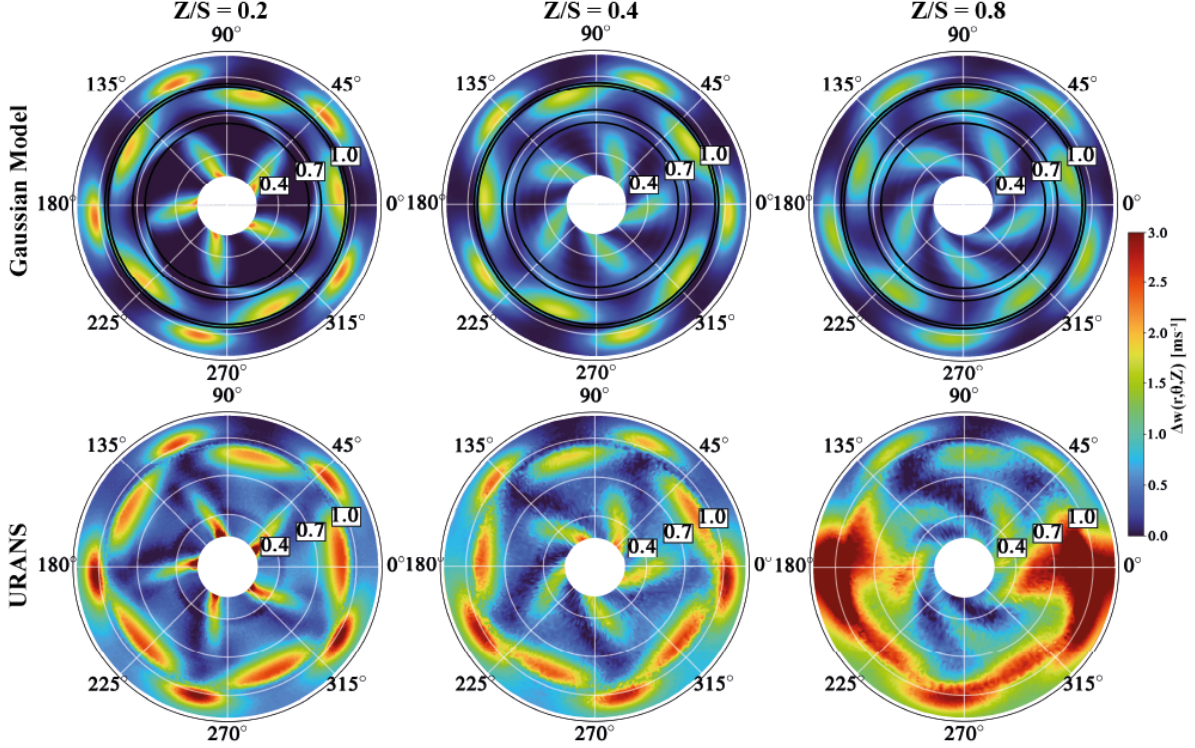


Fig. 13 URANS and Gaussian model velocity deficit for the installed conditions at $Z/S = 0.2, 0.4$, and 0.8 , with black lines indicating interpolated region bounds.

configuration and in the near wake ($Z/S = 0.2$), where blade-to-blade variability and installation effects are negligible, and retains local validity at further downstream locations ($Z/S = 0.4$ and 0.8) only in regions where the flow remains representative of the extracted parameters (e.g., near $\theta \approx 90^\circ$). It cannot, however, capture the global three-dimensional distortion, phase shifts, and azimuthal asymmetry evident in the installed URANS data.

The fitted parameters (A, b, s_0, γ) provide the inputs required by the acoustic formulation of section II. Retaining its amplitude, the axial deficit Δw acts as the deterministic mean wake component; its upwash projection enters Eq. (1), yielding the tonal contribution at multiples of the blade-passing frequency. Normalized by A , the same expression reduces to the modulation function f of Eq. (2), which modulates the homogeneous turbulence field v' and, through the Fourier coefficients H_λ of Eq. (6), supplies the broadband contribution via Eq. (9). The Gaussian reconstruction, therefore, acts as the explicit bridge between the URANS wake field and Amiet’s leading-edge theory, with a single set of fitted parameters governing both the tonal and broadband responses.

V. Conclusions and Future Work

The proposed semi-analytical framework establishes a computationally efficient link between CFD-derived wake characteristics (A, b, s_0, γ) and Amiet’s leading-edge theory, supplying the deterministic and turbulence-modulated inputs required for the tonal and broadband contributions, respectively.

The reconstruction captures the dominant deficit features in the isolated configuration, but in the installed case the wake exhibits asymmetry, deformation, and three-dimensional effects outside the periodic, symmetric formulation. The resulting underprediction of wake width and absence of phase variability are expected to redistribute energy in wavenumber space, affecting both tonal and broadband components by an extent yet to be quantified.

Future work will assess the sensitivity of the acoustic predictions to these wake modeling assumptions and, where necessary, extend the formulation to account for non-periodic, asymmetric, and phase-varying features. These developments are a prerequisite for the framework’s intended application to early-stage eVTOL design exploration, where rapid trade-off studies across propeller geometry, wing placement, and operating conditions would benefit from a tunable parametric wake description.

A. Aeroacoustic Transfer Function

The derivation of the transfer function follows Schwarzschild's theorem for diffraction by a semi-infinite edge [44]. Two regimes are distinguished depending on the gust wavenumber:

- subcritical gust: $\bar{k}_x > \bar{k}_z M / \beta$,
- supercritical gust: $\bar{k}_x < \bar{k}_z M / \beta$,

where all wavenumbers are normalized by the airfoil half-chord $c/2$. The aeroacoustic transfer function used in the broadband noise formulation is defined as

$$L(x, y, z, k_z, k_x) = \int_{-1}^1 g(\xi, k_z, k_x) \exp(-i\mu(M - z/\sigma_0)\xi) d\xi, \quad (13)$$

with $\mu = \bar{k}_z M / \beta^2$ and $\sigma_0 = \sqrt{z^2 + \beta^2(x^2 + y^2)}$. For the supercritical regime, the leading-edge contribution is written as

$$\mathcal{L}_1(x, y, z, k_z, k_x) = \frac{1}{\pi} \sqrt{\frac{2}{(\bar{k}_z + \beta^2 \kappa) \Theta_1}} e^*(2\Theta_1) \exp(i\Theta_2), \quad (14)$$

whereas the corresponding trailing-edge correction term becomes

$$\mathcal{L}_2(x, y, z, k_z, k_x) = \frac{\exp(i\Theta_2)}{\Theta_1 \pi \sqrt{2\pi(\bar{k}_z + \beta^2 \kappa)}} \left\{ i(1 - \exp(-i2\Theta_1)) + (1 - i) \left[e^*(4\kappa) - \sqrt{\frac{2\kappa}{\Theta_3}} \exp(-i2\Theta_1) e^*(2\Theta_3) \right] \right\}, \quad (15)$$

where $\Theta_1 = \kappa - \mu z / \sigma_0$, $\Theta_2 = \mu(M - z/\sigma_0) - \pi/4$, and $\Theta_3 = \kappa + \mu z / \sigma_0$. For subcritical gusts, the leading-edge response becomes

$$\mathcal{L}'_1(x, y, z, k_z, k_x) = \frac{1}{\pi} \sqrt{\frac{2}{(k_z - i\beta^2 \kappa_0) \Theta'_1}} e^*(2\Theta'_1) \exp(i\Theta_2), \quad (16)$$

and the associated trailing-edge correction is approximated by

$$\mathcal{L}'_2(x, y, z, k_z, k_x) \approx \frac{\exp(i\Theta_2)}{\Theta'_1 \pi} \sqrt{\frac{2\pi}{(k_z - i\beta^2 \kappa_0)}} \left\{ i \left(1 - \exp(-i2\Theta'_1) - \operatorname{erf}(\sqrt{4\kappa_0}) \right) + (1 + i) \sqrt{\frac{\kappa_0}{\Theta'_3}} \exp(-i2\Theta'_1) \Phi_0(\sqrt{2i\Theta'_3}) \right\}, \quad (17)$$

with $\Theta'_1 = -i\kappa_0 - \mu z / \sigma_0$, $\Theta_2 = \mu(M - z/\sigma_0) - \pi/4$, $\Theta'_3 = -i\kappa_0 + \mu z / \sigma_0$. The auxiliary function Φ_0 is defined as

$$\Phi_0(Y) = \frac{2}{\sqrt{\pi}} \int_0^Y \exp(-y^2) dy = \frac{\exp(i\pi/4)}{\sqrt{\pi}} \int_0^{-iY^2} \frac{\exp(-y)}{\sqrt{y}} dy. \quad (18)$$

Eq. (14), Eq. (15), Eq. (16), and Eq. (17) correspond respectively to Eqs. (C.47), (C.48), (C.51), and (C.52) in [27].

B. Three-Dimensional Formulation

The wake modulation is extended by introducing an additional harmonic decomposition in the transverse direction with, for instance, $\kappa_x = 2\pi/R_1$, where R_1 is a fictional spanwise period [32]. The resulting modulation function is written as

$$f(z, x, t) = \sum_{\lambda=-\infty}^{\infty} \sum_{\mu=-\infty}^{\infty} H_{\lambda, \mu} \exp \left(i \left(\lambda \left(\kappa_z z - \frac{2\pi t}{T} \right) + \mu \kappa_x x \right) \right). \quad (19)$$

Here, μ represents the transverse harmonic contribution. Applying the double spatial Fourier transform to the modulated velocity field yields

$$\begin{aligned} \hat{v}(k_z, k_x, t) &= \frac{1}{(2\pi)^2} \sum_{\lambda=-\infty}^{\infty} \sum_{\mu=-\infty}^{\infty} H_{\lambda, \mu} \exp \left(-\frac{i2\pi\lambda t}{T} \right) \iint_{-\infty}^{\infty} v'(z, x, t) \exp(i(\lambda\kappa_z z + \mu\kappa_x x)) \exp(-i(k_z z + k_x x)) dz dx \\ &= \sum_{\lambda=-\infty}^{\infty} \sum_{\mu=-\infty}^{\infty} H_{\lambda, \mu} \exp \left(-\frac{i2\pi\lambda t}{T} \right) \hat{v}'(k_z - \lambda\kappa_z, k_x - \mu\kappa_x), \end{aligned} \quad (20)$$

whereas the associated velocity power spectral density becomes

$$\varphi_{vv}(k_z, k_x) = \sum_{\lambda=-\infty}^{\infty} \sum_{\mu=-\infty}^{\infty} |H_{\lambda,\mu}|^2 \varphi_{v'v'}(k_z - \lambda\kappa_z, k_x - \mu\kappa_x). \quad (21)$$

The far-field acoustic response is then obtained by substituting into Amiet's leading-edge theory in Eq. (1):

$$S_{pp}(\mathbf{y}, \omega) = \left(\frac{\rho_0 k x c}{2\sigma_0^2} \right)^2 \pi W \frac{d}{2} |\mathcal{L}(\mathbf{x}, K_z, K_x)|^2 \sum_{\lambda=-\infty}^{\infty} \sum_{\mu=-\infty}^{\infty} |H_{\lambda,\mu}|^2 \varphi_{v'v'}(k_z - \lambda\kappa_z, k_x - \mu\kappa_x). \quad (22)$$

This derivation, adapted from [28], does not support any restrictive assumptions about the distribution of energy in the two-dimensional wavenumber space.

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